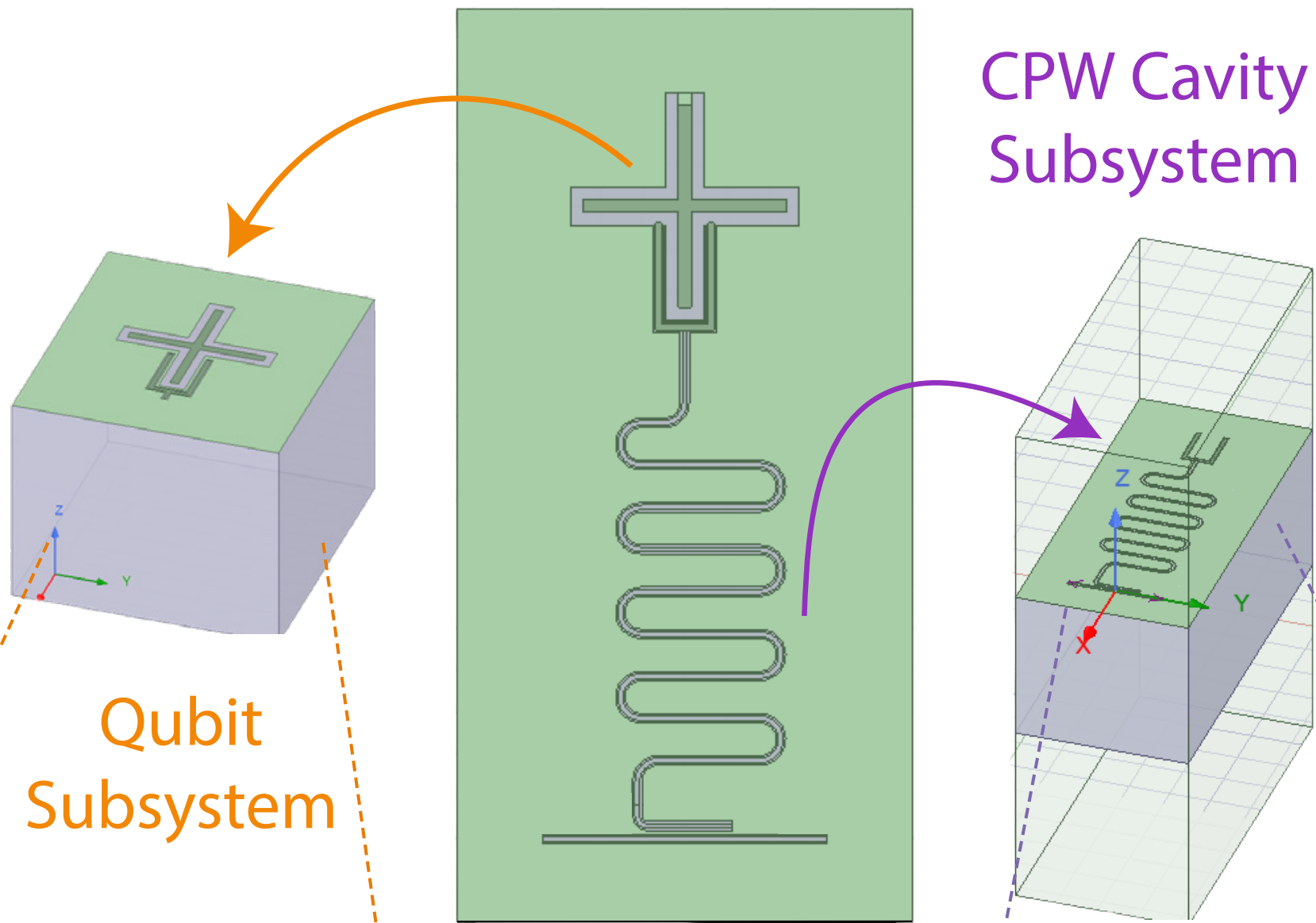




**$\alpha$  (anharmonicity)**

$\approx -E_C$  from Q3D

**$f_{\text{qubit}}$  (qubit frequency)**

$\approx \sqrt{(8 E_J E_C)} - E_C$  from Q3D

**$f_{\text{res}}$  (resonator freq.)**

HFSS eigenmode solver

**$\kappa$  (resonator linewidth)**

HFSS via 50 $\Omega$  port loss